

Notes on Semigroups and Related Structures

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1 Elementary Concepts

1.1 Sets and Relations

1.1.1 Relations

Definition 1.1.1 (Binary relation). Let X be a set. Every subset ρ of the Cartesian product $X \times X$ is called a (*binary*) *relation* on X . If $(x, y) \in \rho$, we also write $x\rho y$, where $x, y \in X$.

Definition 1.1.2 (Domain and range). For a relation $\rho \subseteq X \times X$, the *domain* and *range* of R are defined by

$$\text{dom } \rho = \{x \in X : \exists y \in X \text{ with } (x, y) \in \rho\},$$

and

$$\text{ran } \rho = \{y \in X : \exists x \in X \text{ with } (x, y) \in \rho\}.$$

The word *image* is also used sometimes as a synonym of range.

Notation 1.1.3. The set of all binary relations on X is denoted by $\mathfrak{B}(X)$.

Remark 1.1.4. If $\rho, \tau \in \mathfrak{B}(X)$, then $\rho \subseteq \tau$ means that $x\rho y \Rightarrow x\tau y$ for all $x, y \in X$. We shall often write $\rho \leq \tau$ for $\rho \subseteq \tau$.

Definition 1.1.5 (Complement). The complement of a relation $\rho \in B(X)$ is the relation $\bar{\rho} \in B(X)$ defined by

$$(x, y) \in \bar{\rho} \iff (x, y) \notin \rho.$$

Definition 1.1.6 (Properties of a relation). A relation $\rho \in (X)$ is called

1. *reflexive*, if $x\rho x$ for all $x \in X$;
2. *symmetric*, if $x\rho y$ implies $y\rho x$;
3. *antisymmetric*, if $x\rho y$ and $y\rho x$ together imply $x = y$;
4. *transitive*, if $x\rho y$ and $y\rho z$ imply $x\rho z$;
5. *right unique*, if ρy and $x\rho z$ imply $y = z$;
6. *left unique*, if $x\rho y$ and $z\rho y$ imply $x = z$;
7. *right total*, if for every $y \in X$ there exists $x \in X$ such that $x\rho y$;
8. *left total*, if for every $x \in X$ there exists $y \in X$ such that $x\rho y$.

The last four notions also apply to relations $\rho \subseteq X \times Y$ for two possibly different sets X and Y .

1.1.2 Mappings

Definition 1.1.7 (Mapping). Let X and Y be sets. A mapping $\varphi : X \rightarrow Y$ is a relation $\varphi \subseteq X \times Y$ which is right unique and left total; that is, for every $x \in X$ there exists exactly one $y \in Y$ such that $(x, y) \in \varphi$.

Remark 1.1.8. As usual, we write $\varphi(x) = y$. Sometimes we also write $x\varphi = y$. The set of all mappings from X to Y is denoted by $\text{Map}(X, Y)$ or Y^X . We also use $\mathcal{T}(X)$ for $\text{Map}(X, X)$. The elements of $\mathcal{T}(X)$ are called *transformations* of X .

Definition 1.1.9 (Injective, surjective, bijective). A mapping $\varphi : X \rightarrow Y$ is called

1. *injective*, if $\varphi(x) = \varphi(x') \Rightarrow x = x'$ for all $x, x' \in X$;
2. *surjective*, if for every $y \in Y$ there exists $x \in X$ such that $\varphi(x) = y$;
3. *bijective*, if it is both injective and surjective.

Equivalently, injectivity means that the relation is left unique, and surjectivity means that the relation is right total.

Remark 1.1.10. The subset of all bijective transformations in $\mathcal{T}(X)$ is denoted by $\mathcal{S}(X)$, and its elements are called *permutations* or *bijections* of X .

Definition 1.1.11 (Identity map). The identity map $\text{id}_X : X \rightarrow X$ is defined by

$$\text{id}_X(x) = x \quad \forall x \in X.$$

Definition 1.1.12 (Constant mapping). For $y \in Y$, the constant mapping $c_y : X \rightarrow Y$ is defined by

$$c_y(x) = y \quad \forall x \in X.$$

Definition 1.1.13 (Restriction). If $\varphi : X \rightarrow Y$ is a mapping and $A \subseteq X$, then the restriction of φ to A is defined by $\varphi|_A : A \rightarrow Y$ such that

$$\varphi|_A(a) = \varphi(a) \quad \forall a \in A.$$

Conversely, we say that φ extends $\varphi|_A$.

Definition 1.1.14 (Partial mapping). A partial mapping $\varphi : X \rightarrow Y$ is defined to be a right unique relation; that is, $\varphi : \text{dom } \varphi \rightarrow Y$ is a mapping, where $\text{dom } \varphi \subseteq X$. The set $\text{dom } \varphi$ is called the domain of φ . The set of all partial mappings on X (from X to X) is denoted by $\mathcal{PT}(X)$.

Remark 1.1.15. The mapping $\varphi : \text{dom } \varphi \rightarrow \text{im } \varphi$ with $\text{dom } \varphi = \text{im } \varphi = \emptyset$ is called the empty mapping.

1.1.3 Order Relations

Definition 1.1.16 (Partial order). A reflexive, antisymmetric, and transitive relation α on a set X is called a *partial ordering* on X , or simply an ordering of X . One writes $x \leq y$ for $(x, y) \in \alpha$, and the pair $(X, \leq)$ is called a *partially ordered set* or a *poset*.

Definition 1.1.17 (Total order). If in addition, a partial ordering has the property that for every $x, y \in X$, either $x \leq y$ or $y \leq x$. In this case, $\leq$ is called a *total order*. Then $(X, \leq)$ is also called a *linearly ordered set* or a *chain*.

Definition 1.1.18 (Maximal and minimal elements). Let X be a poset. An element $w \in X$ is called a *maximal* (respectively, *minimal*) element of $(X, \leq)$ if $w \leq u$ (respectively, $u \leq w$) implies $u = w$ for every $u \in X$.

Definition 1.1.19 (Minimal condition and well-ordering). A poset $(X, \leq)$ satisfies the *minimal condition* if every non-empty subset of X has a minimal element. In that case, we also say that $(X, \leq)$ is *well ordered*.

Definition 1.1.20 (Lower bound and meet). Let Y be a non-empty subset of a poset X . An element $c \in X$ is called a *lower bound* for Y if $c \leq y$ for all $y \in Y$. An element $u \in X$ is called the *greatest lower bound* or *meet* of Y if u is a lower bound for Y and $z \leq u$ for every lower bound z of Y . The greatest lower bound of Y is denoted by $\bigwedge y : y \in Y$. If $Y = \{a, b\}$, we also write $a \wedge b$.

Definition 1.1.21 (Lower semilattice). If $(X, \leq)$ is such that $a \wedge b$ exists for every pair $a, b \in X$, then $(X, \leq)$ is called a *lower semilattice*. If, in addition, $\bigwedge \{y : y \in Y\}$ exists for every non-empty subset $Y \subseteq X$, then $(X, \leq)$ is called a *complete lower semilattice*.

Definition 1.1.22 (Upper bound and join). Similarly, for a non-empty subset $Y \subseteq X$, an element $c \in X$ is called an *upper bound* for Y if $y \leq c$ for all $y \in Y$. An element $v \in X$ is called the *least upper bound* or *join* of Y if v is an upper bound for Y and $v \leq z$ for every upper bound z of Y . The least upper bound of Y is denoted by $\bigvee \{y : y \in Y\}$. If $Y = \{a, b\}$, we also write $a \vee b$.

Definition 1.1.23 (Upper semilattice). If $(X, \leq)$ is such that $a \vee b$ exists for every pair $a, b \in X$, then $(X, \leq)$ is called an *upper semilattice*. If, in addition, $\bigvee y : y \in Y$ exists for every non-empty subset $Y \subseteq X$, then $(X, \leq)$ is called a *complete upper semilattice*.

Definition 1.1.24 (Lattice). If a poset is both a (complete) lower semilattice and a (complete) upper semilattice then it is called a (complete) lattice. In this case we use the notation $(X, \leq, \wedge, \vee)$.

Remark 1.1.25. The set $\mathfrak{B}(X)$ of all binary relations on a set X is a poset under inclusion; that is, $\rho \leq \tau$ if and only if $\rho \subseteq \tau$, for $\rho, \tau \in \mathfrak{B}(X)$. Moreover, it is a lattice with

$$\rho \wedge \tau = \rho \cap \tau \quad \text{and} \quad \rho \vee \tau = \rho \cup \tau,$$

which is called the lattice of relations on X and is denoted by $(\mathfrak{B}(X), \leq, \wedge, \vee)$.

Lemma 1.1.1 (Zorn's lemma). If $(X, \leq)$ is a non-empty poset such that every chain of elements has an upper bound in X , then X has at least one maximal element.

Theorem 1.1.2 (Zermelo's theorem on well-ordering). Every non-empty set can be well ordered.

1.1.4 Equivalence relations and partitions

Definition 1.1.26. An *equivalence* or *equivalence relation* on X is a reflexive, symmetric and transitive relation on X .

Special equivalence relations on X are $\text{id}_X = \{(x, x) \mid x \in X\}$, the *diagonal relation* on X , also denoted by Δ_X , and the *total relation* $X \times X$, also denoted by ∇_X .

Definition 1.1.27. Let $K \subseteq X$ be a subset. Define the *Rees equivalence* ρ_K by setting

$$x \rho_K x' \iff x, x' \in K \text{ or } x = x'.$$

We denote the set of all equivalences on a set X by $\mathcal{E}(X)$. As a subset of $\mathfrak{B}(X)$ it is partially ordered by inclusion.

We call the intersection of all equivalences containing the relation $\rho \in \mathcal{B}(X)$ the *equivalence on X generated by ρ* or the *equivalence closure* of ρ , and denote it by ρ^e .

If $\{\rho_i \in \mathcal{E}(X) \mid i \in I\}$ is a non-empty set of equivalences on X , then $\bigcap_{i \in I} \rho_i$ is again an equivalence on X .

If $\rho \in \mathcal{B}(X)$, then the intersection of all equivalences containing ρ is the unique minimal equivalence on X containing ρ .

In fact $\mathcal{E}(X)$ is a lattice. If $\rho, \sigma \in \mathcal{E}(X)$, then their greatest lower bound $\rho \wedge \sigma$ is $\rho \cap \sigma$ and their least upper bound $\rho \vee \sigma$ is $(\rho \cup \sigma)^e$.

Definition 1.1.28. For $\rho \in \mathcal{E}(X)$ and $A \subseteq X$, set

$$\rho(A) := \{b \in X \mid \text{there exists } a \in A \text{ with } (a, b) \in \rho\}.$$

For $A = \{a\}$ we write instead of $\rho(a)$ also $[a]_\rho$ or just $[a]$, and call this set the ρ -class of a or the class of a (with respect to ρ).

Let $\rho, \sigma \in \mathcal{E}(X)$. Then $\rho \subseteq \sigma$ if and only if every σ -class is a union of ρ -classes.

Definition 1.1.29. A family of subsets $X_i, i \in I$, of a set X is called a *partition* of X if the subsets X_i are pairwise disjoint, non-empty, and their union is X .

If ρ is an equivalence relation on X , then the set of equivalence classes

$$\pi(\rho) = \{[x]_\rho \mid x \in X\}$$

is the partition of X corresponding to ρ .

On the other hand, if $\pi = \{X_i \mid i \in I\}$ is a partition of X into subsets X_i , then the relation

$$\rho(\pi) = \{(x, y) \in X \times X \mid \text{there exists } i \in I \text{ such that } x, y \in X_i\}$$

is an equivalence relation on X . Note that

$$\rho(\pi(\rho)) = \rho, \quad \pi(\rho(\pi)) = \pi.$$

The set of classes with respect to an equivalence relation ρ is called the *factor set* of X with respect to ρ and denoted by X/ρ . The mapping

$$\pi_\rho : X \rightarrow X/\rho, \quad x \mapsto [x]_\rho$$

is called the *canonical surjection* with respect to ρ .

For any $\rho \in \mathfrak{B}(X)$, let

$$\begin{aligned} \rho^R &= \bigcap \{\sigma \in \mathfrak{B}(X) \mid \rho \subseteq \sigma, \sigma \text{ is reflexive}\}, \\ \rho^S &= \bigcap \{\sigma \in \mathfrak{B}(X) \mid \rho \subseteq \sigma, \sigma \text{ is symmetric}\}. \end{aligned}$$

There is at least one element $\sigma \in \mathfrak{B}(X)$ satisfying the condition in each of the collections above, namely $\sigma = X \times X$. Furthermore, since every such σ contains ρ , the intersections ρ^R and ρ^S are both non-empty. It is easy to see that

- ρ^R , called the *reflexive closure* of ρ , is the smallest reflexive relation containing ρ ;
- ρ^S , called the *symmetric closure* of ρ , is the smallest symmetric relation containing ρ .

1.1.5 Inverse relation, restriction and composition

Let $\rho \subseteq X \times Y$. The *inverse* (*converse*) relation to ρ is

$$\rho^{-1} = \{(y, x) \in Y \times X \mid (x, y) \in \rho\}.$$

Similar as for mappings, we now define the *restriction* $\rho|_{X'}$ of a relation $\rho \in B(X)$ for a subset $X' \subseteq X$ by

$$\rho|_{X'} = \rho \cap (X' \times X').$$

The *composition* of relations $\rho, \sigma \in \mathfrak{B}(X)$ is defined as

$$\rho \circ_r \sigma = \{(x, z) \in X \times X \mid \text{there exists } y \in X \text{ with } x\sigma y \text{ and } y\rho z\},$$

if we start with the right relation σ , and analogously for $\rho \subseteq X \times Y, \sigma \subseteq Y \times Z$ as

$$\rho \circ_r \sigma = \{(x, z) \in X \times Z \mid \text{there exists } y \in Y \text{ with } x\sigma y \text{ and } y\rho z\}.$$

If we start with the left relation ρ , we define

$$\rho \circ_l \sigma = \{(x, z) \in X \times X \mid \text{there exists } y \in X \text{ with } x\rho y \text{ and } y\sigma z\},$$

for $\rho, \sigma \in B(X)$, and analogously for $\rho \subseteq X \times Y, \sigma \subseteq Y \times Z$ as

$$\rho \circ_l \sigma = \{(x, z) \in X \times Z \mid \text{there exists } y \in Y \text{ with } x\rho y \text{ and } y\sigma z\}.$$

Usually, for convenience, we will denote either $\circ_r$ or $\circ_l$ by $\circ$. We will be careful in any situation to indicate the use of either convention.

If $\varphi \in \text{Map}(X, Y)$ and $\psi \in \text{Map}(Y, Z)$, then in the case when mapping symbols are written to the left of the elements on which they act, by $\psi\varphi$ we mean $\psi \circ_r \varphi$, i.e.

$$(\psi\varphi)(x) = \psi(\varphi(x)) \quad \forall x \in X.$$

In the case when mapping symbols are written to the right of the elements on which they act, by $x\varphi\psi$ we mean $\varphi \circ_l \psi$, i.e.

$$(x\varphi)\psi = (x\varphi)\psi \quad \forall x \in X.$$

1.1.6 Transitive and equivalence closure

It is clear that the operation $\circ$ on $\mathfrak{B}(X)$ is associative and thus $(\mathfrak{B}(X), \circ)$ will be one of the examples of monoids (definition in a further section) with the identity element id_X . Sometimes we write $\rho\sigma$ for $\rho \circ \sigma$. We define powers of a relation $\rho \in \mathfrak{B}(X)$ as follows:

$$\rho^0 = id_X, \quad \rho^i = \rho \circ \rho \circ \dots \circ \rho, \text{ } i \text{ times}, \quad i \in \mathbb{N}.$$

Definition 1.1.30. The *transitive closure* of a relation $\rho \in \mathfrak{B}(X)$ is the relation

$$\rho^\infty = \bigcup_{n=1}^{\infty} \rho^n \in \mathfrak{B}(X).$$

Lemma 1.1.3. If $\rho \in B(X)$, then ρ^∞ is the smallest transitive relation in $(B(X), \leq)$ containing ρ .

Proof. If $(x, y), (y, z) \in \rho^\infty$, then there exist $m, n \in \mathbb{N}$ such that $(x, y) \in \rho^m, (y, z) \in \rho^n$, and

$$(x, z) \in \rho^m \circ \rho^n = \rho^{m+n} \subseteq \rho^\infty.$$

Thus ρ^∞ is a transitive relation and $\rho \subseteq \rho^\infty$. If now σ is a transitive relation on X and $\rho \subseteq \sigma$, then

$$\rho^2 = \rho \circ \rho \subseteq \sigma \circ \sigma \subseteq \sigma.$$

By induction it follows easily that $\rho^n \subseteq \sigma$ for all $n \in \mathbb{N}$, i.e. $\rho^\infty \subseteq \sigma$. □

Theorem 1.1.4. If $\rho \in \mathfrak{B}(X)$, then for the equivalence closure ρ^e we get

$$\rho^e = (\rho \cup \rho^{-1} \cup id_X)^\infty.$$

Proof. It is clear that $\tau = (\rho \cup \rho^{-1} \cup id_X)^\infty$ contains ρ and is transitive and reflexive. The relation $\nu = \rho \cup \rho^{-1} \cup id_X$ is symmetric and therefore $\nu^n = (\nu^{-1})^n = (\nu^n)^{-1}$. Thus, if $(x, y) \in \tau$, then $(x, y) \in \nu^n$ for some $n \in \mathbb{N}$ and therefore $(y, x) \in \nu^n$. Thus $(y, x) \in \tau$, i.e. τ is symmetric. This means that $\tau \in \mathcal{E}(X)$.

If $\sigma \in \mathcal{E}(X)$ and $\rho \subseteq \sigma$, then $id_X \subseteq \sigma, \rho^{-1} \subseteq \sigma^{-1} = \sigma$, and therefore $\nu = \rho \cup \rho^{-1} \cup id_X \subseteq \sigma$. Since $\nu \circ \nu \subseteq \sigma \circ \sigma \subseteq \sigma$, we get $\nu^n \subseteq \sigma$ for all $n \in \mathbb{N}$, and hence $\tau \subseteq \sigma$. □

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Corollary 1.1.5. If $\rho \in \mathfrak{B}(X)$, then $(x, y) \in \rho^e$ if and only if $x = y$ or for some $n \in \mathbb{N}$ there exists a sequence of elements $x_0, x_1, x_2, \dots, x_n$ in X such that $x_0 = x$, $x_n = y$, and for every $i \in \{1, \dots, n\}$ either $(x_{i-1}, x_i) \in \rho$ or $(x_i, x_{i-1}) \in \rho$.

In particular, if ρ and σ are equivalence relations on a set X , then in $\mathcal{E}(X)$ their join $\rho \vee \sigma$ is the relation defined by

$$x(\rho \vee \sigma)y \iff \text{there exist } x_0, x_1, \dots, x_n \in X \text{ such that } x = x_0, y = x_n$$

and

$$(x_{i-1}, x_i) \in \tau_i, \quad \tau_i \in \{\rho, \sigma\}, \quad i \in \{1, \dots, n\}.$$

Proposition 1.1.6. For any $\rho \in \mathfrak{B}(X)$,

1. $\rho^R = \rho \cup id_X$;
2. $\rho^S = \rho \cup \rho^{-1}$;
3. $(\rho^R)^S = (\rho^S)^R = \rho \cup \rho^{-1} \cup id_X$;
4. $\rho^e = \left((\rho^R)^S \right)^\infty$.

1.1.7 Kernel equivalence and Homomorphism Theorem for sets

Definition 1.1.31. Let $\varphi : X \rightarrow Y$ be a mapping. The relation

$$\ker \varphi = \{(x, x') \in X \times X \mid \varphi(x) = \varphi(x')\}$$

is an equivalence on X which is called the *kernel equivalence* or the *kernel* of φ .

It follows from the definition of the composition of relations that

$$\ker \varphi = \varphi^{-1} \circ \varphi = \varphi^{-1} \circ_r \varphi.$$

Theorem 1.1.7 (Homomorphism Theorem for sets). Let $\varphi : X \rightarrow Y$ be a mapping and let ρ be an equivalence on X such that $x\rho x'$ implies $\varphi(x) = \varphi(x')$, i.e. $\rho \leq \ker \varphi$. Then $\varphi' : X/\rho \rightarrow Y$, with $\varphi'([x]_\rho) := \varphi(x)$, $x \in X$, is the unique mapping such that the following diagram is commutative:

$$\begin{array}{ccc} X & \xrightarrow{\varphi} & Y \\ \pi_\rho \downarrow & \nearrow \varphi' & \\ X/\rho & & \end{array}$$

i.e. such that $\varphi' \pi_\rho = \varphi$, where π_ρ denotes the canonical surjection.

If $\rho = \ker \varphi$, then φ' is injective, and if φ is surjective, then φ' is surjective.

Proof. Define $\varphi' : X/\rho \rightarrow Y$ by $\varphi'([x]_\rho) = \varphi(x)$ for any $x \in X$. Suppose $[x]_\rho = [x']_\rho$ for $x, x' \in X$. Then $(x, x') \in \rho$ and thus $\varphi(x) = \varphi(x')$. Hence φ' is well-defined.

Suppose $\varphi'([x]_\rho) = \varphi'([x']_\rho)$. Then $\varphi(x) = \varphi(x')$ and thus $(x, x') \in \ker \varphi$.

If $\rho = \ker \varphi$, then $[x]_\rho = [x']_\rho$ and thus φ' is injective. □

1.1.8 Congruences

Definition 1.1.32. A relation ρ on X is called

- *left compatible* if for all $x, y, z \in X$,

$$(x, y) \in \rho \implies (zx, zy) \in \rho;$$

- *right compatible* if for all $x, y, z \in X$,

$$(x, y) \in \rho \implies (xz, yz) \in \rho;$$

- *compatible* if for all $x, y, s, t \in X$,

$$(x, y) \in \rho \text{ and } (s, t) \in \rho \implies (xs, yt) \in \rho.$$

A left compatible equivalence relation is called a *left congruence*; a right compatible equivalence relation is called a *right congruence*; and a compatible equivalence relation is called a *congruence*.

Proposition 1.1.8. *A relation ρ on X is a congruence if and only if it is both a left and a right congruence.*

Proof. Suppose ρ is both a left and a right congruence. Let $x, y, z, t \in X$ be such that $(x, y) \in \rho$ and $(z, t) \in \rho$. Since ρ is a right congruence, $(xz, yz) \in \rho$, and since ρ is a left congruence, $(yz, yt) \in \rho$. By transitivity of ρ , it follows that $(xz, yt) \in \rho$. Hence ρ is a congruence.

Conversely, suppose ρ is a congruence. Let $x, y \in X$ be such that $(x, y) \in \rho$, and let $z \in X$. Since ρ is reflexive, $(z, z) \in \rho$. By compatibility, $(zx, zy) \in \rho$. Thus ρ is left compatible. Similarly, $(xz, yz) \in \rho$, so ρ is right compatible. Hence ρ is both a left and a right congruence. $\square$

For any $\rho \in \mathfrak{B}(X)$, let

$$\rho^c = \bigcap \{ \sigma \in \mathfrak{B}(X) \mid \rho \subseteq \sigma, \sigma \text{ is left and right compatible} \}.$$

Then ρ^c is the smallest left and right compatible relation containing ρ . Also, let

$$\rho^\# = \bigcap \{ \sigma \in \mathfrak{B}(X) \mid \rho \subseteq \sigma, \sigma \text{ is a congruence} \}.$$

Then ρ^c , called the congruence generated by ρ , is the smallest congruence containing ρ .

Proposition 1.1.9. *For any $\rho, \sigma \in \mathfrak{B}(X)$,*

1. $(\rho \cup \sigma)^c = \rho^c \cup \sigma^c$;
2. $(\rho^{-1})^c = (\rho^c)^{-1}$.

Proposition 1.1.10. *For any $\rho \in \mathfrak{B}(X)$, we have*

$$\rho^\# = (\rho^c)^e.$$

Proof. By definition, $(\rho^c)^e$ is an equivalence relation containing ρ^c , and hence containing ρ . Notice that

$$\begin{aligned} (\rho^c)^e &= (\rho^c \cup (\rho^c)^{-1} \cup id_X)^\infty \\ &= ((\rho \cup \rho^{-1} \cup id_X)^c)^\infty \\ &= \bigcup_{n=0}^\infty ((\rho \cup \rho^{-1} \cup id_X)^c)^n \end{aligned}$$

Now, for all $x, y, z \in X$,

$$\begin{aligned} &(x, y) \in (\rho^c)^e \\ \implies &(x, y) \in \bigcup_{n=0}^\infty ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &\exists n \in \mathbb{N} \text{ such that } (x, y) \in ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &\exists x_0, x_1, \dots, x_n \text{ such that } x = x_0 \text{ and } y = x_n \text{ and } (\forall i) (x_i, x_{i+1}) \in (\rho \cup \rho^{-1} \cup id_X)^c \\ \implies &zx = zx_0, zx_n = zy \text{ and } (\forall i) (zx_i, zx_{i+1}) \in (\rho \cup \rho^{-1} \cup id_X)^c \\ &\quad (\text{since } (\rho \cup \rho^{-1} \cup id_X)^c \text{ is left and right congruence}) \\ \implies &(zx, zy) \in ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &(zx, zy) \in \bigcup_{n=0}^\infty ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &(zx, zy) \in (\rho^c)^e. \end{aligned}$$

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Therefore, $(\rho^c)^e$ is left compatible. Similarly, $(\rho^c)^e$ is right compatible. Hence $(\rho^c)^e$ is a congruence containing ρ .

Finally, let τ be any congruence containing ρ . Then τ is left and right compatible, so $\rho^c \subseteq \tau$. Since τ is an equivalence relation, it must contain $(\rho^c)^e$. Hence $(\rho^c)^e \subseteq \tau$. Therefore, $(\rho^c)^e$ is the smallest congruence containing ρ , and therefore $\rho^\# = (\rho^c)^e$. Consequently, $\rho^\# = (\rho^c)^e$ $\square$

We denote by $\mathcal{C}(X)$ the set of all congruences on X . Then $\mathcal{C}(X)$, as a subset of $\mathcal{C}(X)$ and $\mathfrak{B}(X)$, is partially ordered by inclusion.

In fact, $\mathcal{C}(X)$ forms a lattice. For $\rho, \sigma \in \mathcal{C}(X)$, we have

$$\rho \wedge \sigma = \rho \cap \sigma \quad \text{and} \quad \rho \vee \sigma = (\rho \cup \sigma)^\#.$$

Also,

$$\begin{aligned} (\rho \cup \sigma)^\# &= ((\rho \cup \sigma)^c)^e \\ &= (\rho^c \cup \sigma^c)^e \\ &= (\rho \cup \sigma)^e \end{aligned}$$

so there is no ambiguity in writing $\rho \vee \sigma$ whether one means the join $(\rho \vee \sigma)^e$ in the lattice $\mathcal{C}(X)$ or the join $(\rho \vee \sigma)^\#$ in the lattice $\mathcal{C}(X)$.

Proposition 1.1.11. *Let $\rho, \sigma \in \mathcal{C}(X)$. Then*

$$\rho \vee \sigma = (\rho \circ \sigma)^\infty.$$

Proof. Let $\rho, \sigma \in \mathcal{C}(X)$. Since $\rho \vee \sigma$ contains both ρ and σ , it follows that

$$\rho \circ \sigma \subseteq (\rho \cup \sigma) \circ (\rho \cup \sigma) = (\rho \vee \sigma)^2.$$

More generally, $(\rho \circ \sigma)^n \subseteq (\rho \cup \sigma)^{2n}$. Thus,

$$(\rho \circ \sigma)^\infty = \bigcup_{n=0}^{\infty} (\rho \circ \sigma)^n \subseteq \bigcup_{n=0}^{\infty} (\rho \cup \sigma)^{2n} = (\rho \cup \sigma)^\infty.$$

On the other hand, $\rho \circ \sigma \supseteq \rho \circ id_X = \rho$ (since σ is reflexive), and $\rho \circ \sigma \supseteq id_X \circ \sigma = \sigma$ (since ρ is reflexive). Thus, $\rho \cup \sigma \subseteq \rho \circ \sigma$. Hence, $(\rho \cup \sigma)^\infty \subseteq (\rho \circ \sigma)^\infty$.

Consequently, $(\rho \cup \sigma)^\infty = (\rho \circ \sigma)^\infty$.

Then,

$$\begin{aligned} \rho \vee \sigma &= (\rho \cup \sigma)^e = \left(((\rho \cup \sigma)^R)^S \right)^\infty \\ &= ((\rho \cup \sigma) \cup (\rho \cup \sigma)^{-1} \cup id_X)^\infty \\ &= (\rho \cup \sigma \cup \rho^{-1} \cup \sigma^{-1} \cup id_X)^\infty \\ &= (\rho \cup \sigma \cup id_X)^\infty \\ &= (\rho \cup \sigma)^\infty = (\rho \circ \sigma)^\infty. \end{aligned}$$

This completes the proof. $\square$

Proposition 1.1.12. *Let $\rho, \sigma \in \mathcal{C}(X)$. If $\rho \circ \sigma = \sigma \circ \rho$, then*

$$\rho \vee \sigma = \rho \circ \sigma.$$

Proof. Suppose $\rho \circ \sigma = \sigma \circ \rho$. Then

$$(\rho \circ \sigma)^2 = \rho \circ \sigma \circ \rho \circ \sigma = \rho \circ \rho \circ \sigma \circ \sigma = \rho \circ \sigma,$$

since ρ and σ are transitive.

Hence, $(\rho \circ \sigma)^n = \rho \circ \sigma$ for all $n \in \mathbb{N}$, and therefore

$$\rho \circ \sigma = (\rho \circ \sigma)^\infty = \bigcup_{n \geq 0} (\rho \circ \sigma)^n = \rho \circ \sigma.$$

So $\rho \vee \sigma = \rho \circ \sigma$. $\square$

1.2 Introductory Algebraic structures

1.2.1 Groupoids, Identities, and Zeros

Definition 1.2.1. A *groupoid* is a non-empty set A together with a binary operation

$$A \times A \rightarrow A, \quad (a, a') \mapsto aa'.$$

The operation is usually called the *multiplication* of the groupoid.

Remark 1.2.2. Let A be a groupoid. For non-empty subsets $B, C \subseteq A$, define

$$BC = \{bc \mid b \in B, c \in C\}.$$

We write $B^2 = BB$, and similarly for higher products.

Definition 1.2.3. Let S be a groupoid. An element $e \in S$ is called

- a *left identity* if $es = s$ for all $s \in S$;
- a *right identity* if $se = s$ for all $s \in S$;
- an *identity* if $es = s = se$ for all $s \in S$.

An identity is also called a *neutral element*.

Proposition 1.2.1. If a groupoid S has both a left and a right identity, then they coincide. Hence a groupoid has at most one identity.

Proof. Let e' be a left identity and e'' a right identity. Then

$$e'' = e'e'' = e'.$$

□

We denote the identity (when it exists) by 1 or more precisely by 1_S .

Definition 1.2.4. Let S be a groupoid with identity.

- $s \in S$ is *left invertible* if there exists $t \in S$ such that $ts = 1$.
- s is *right invertible* if there exists $t \in S$ such that $st = 1$.
- s is *invertible* if there exists $t \in S$ satisfying $ts = st = 1$.

In third case, such an element t is called an inverse of s , and is denoted by s^{-1} whenever it is unique.

Definition 1.2.5. Let S be a groupoid. An element $z \in S$ is called

- a *left zero* if $zs = z$ for every $s \in S$;
- a *right zero* if $sz = z$ for every $s \in S$;
- a *zero* if $zs = z = sz$ for every $s \in S$.

A zero is usually denoted by 0.

Exercise 1.2.2. Show that if a groupoid has both a left zero and a right zero, then they coincide. Hence a groupoid has at most one zero.

1.2.2 Semigroups, monoids and groups

Definition 1.2.6. A *semigroup* is a groupoid whose multiplication is associative. A semigroup with an identity element is called a *monoid*.

Proposition 1.2.3. *If an element of a monoid has both a left and a right inverse, then they coincide.*

Proof. Suppose $ts = 1$ and $st' = 1$. Then

$$t' = (ts)t' = t(st') = t. \quad \square$$

Corollary 1.2.4. *Every element of a monoid has at most one inverse.*

Definition 1.2.7. A *group* G is a monoid in which every element is invertible.

For any $a, b \in G$,

$$(ab)^{-1} = b^{-1}a^{-1}, \quad (a^{-1})^{-1} = a.$$

Exercise 1.2.5. *Suppose S is a semigroup with an element e satisfying $es = s$ for every $s \in S$, and suppose every $s \in S$ has an element $t \in S$ such that $ts = e$. Show that S is a group.*

Remark 1.2.8 (Free semigroup and free monoid). Let X be a non-empty set, called an *alphabet*. Define

$$X^+ = \{x_1x_2 \cdots x_n \mid n \in \mathbb{N}, x_i \in X, 1 \leq i \leq n\},$$

the set of all finite non-empty words over X .

Two words

$$x_1x_2 \cdots x_n, \quad y_1y_2 \cdots y_m$$

are equal if and only if

$$n = m \quad \text{and} \quad x_i = y_i \quad \text{for all } 1 \leq i \leq n.$$

Define the multiplication (concatenation) of words by

$$(x_1x_2 \cdots x_n)(y_1y_2 \cdots y_m) = x_1x_2 \cdots x_ny_1y_2 \cdots y_m.$$

With this operation, X^+ forms a semigroup, called the *free semigroup* on X , or the *free semigroup with basis X* .

Now let

$$X^* = X^+ \cup \{\varepsilon\},$$

where ε denotes the empty word. Then X^* is a monoid under concatenation, with identity element ε . It is called the *free monoid* on X , or the *free monoid with basis X* .

1.2.3 Examples

Example 1.2.9 (Groupoids). 1. $(\mathbb{Z}, -)$.

2. $(\mathbb{Q} \setminus \{0\}, /)$.

3. $(\mathbb{N}, \uparrow)$, where $n \uparrow m = n^m$.

4. The vector product $(\mathbb{R}^3, \times)$.

Example 1.2.10 (Semigroups and monoids). 1. If $(R, +, \cdot)$ is a ring, then $(R, +)$ is a group and $(R, \cdot)$ is a semigroup.

2. $(\mathbb{N}, \cdot)$ is a monoid.

3. $(\mathbb{N}, +)$ is a semigroup.

4. $(\mathcal{T}(X), \circ)$, the full transformation monoid.
5. $(\mathcal{PT}(X), \circ)$, the monoid of partial transformations.
6. $(\mathfrak{B}(X), \circ)$, the semigroup of binary relations.
7. The free semigroup X^+ .
8. The free monoid X^* .
9. $(\mathbb{Z}, \gcd)$.
10. **The bicyclic monoid:**
Let $B = \mathbb{N}_0 \times \mathbb{N}_0$, where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. Define a binary operation on B by

$$(m, n)(p, q) = (m - n + \max\{n, p\}, q - p + \max\{n, p\}).$$

Equivalently,

$$(m, n)(p, q) = \begin{cases} (m - n + p, q), & \text{if } p \geq n, \\ (m, n - p + q), & \text{if } p \leq n. \end{cases}$$

Then $(B, \cdot)$ is a monoid, called the *bicyclic monoid*, with identity element $(0, 0)$.

Example 1.2.11 (Groups). 1. $(\mathbb{Z}, +)$.

2. $(\mathbb{Q}, +)$.
3. $(\mathbb{Q} \setminus \{0\}, \cdot)$.
4. The residue class group $(\mathbb{Z}_r, +)$.
5. The symmetric group $S(X)$, or S_n when $|X| = n$.

2 Semigroups